RAMCO INSTITUTE OF TECHNOLOGY, RAJAPALAYAM

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS3711 - SUMMER INTERNSHIP

INTERNSHIP REPORT

WELLHEAD PRESSURE MONITORING DASHBOARD

Submitted by

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BONAFIDE CERTIFICATE

Certified that this internship report entitled "**WELLHEAD PRESSURE MONITORING DASHBOARD**" is the bona fide work of "**SELVAMANI M (953623104098)**" who carried out the internship at **ECRIO INDIA PVT LTD** under the supervision of **Mr.Vivek.V AP/CSE– RIT.**

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DECLARATION

I hereby declare that this report presents the work carried out by me during the Summer Internship at **ECRIO INDIA PVT LTD** for the specified duration and that the information presented is true to the best of my knowledge.

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EXECUTIVE SUMMARY

The internship project focused on the development of a **Wellhead Pressure Monitoring Dashboard** for continuous monitoring of pressure conditions across oil and gas wells. The project is designed to provide operations and well-integrity engineers with a single live view of tubing, casing, annulus, flowline, and choke pressures across the well stock. The primary objective is to improve well surveillance, identify abnormal pressure behaviour at an early stage, and reduce dependence on manual well-to-well gauge reading. The proposed system collects pressure measurements from pressure transmitters installed at the tubing head, accessible casing annuli such as A, B, and C, flowline, and choke. These measurements are acquired through wellhead Remote Terminal Units (RTUs) at intervals of approximately 1–10 seconds and are published as normalized, well-tagged readings. The monitoring workflow includes validation of sensor values for acceptable range, calibration drift, and stuck-sensor conditions. Instrument-related problems are maintained as a separate alarm class so that sensor faults can be distinguished from actual well-integrity events.

A per-well YAML configuration register is used to maintain important well-specific information such as completion type, available annuli, Maximum Allowable Annulus Surface Pressure (MAASP) for each annulus, tubing Maximum Allowable Operating Pressure (MAOP), normal operating pressure bands, and expected shut-in pressure. These configuration values are used by the rule engine to evaluate incoming pressure measurements and identify abnormal operating conditions. The system implements rule-based monitoring for tubing high and high-high pressure conditions, low and low-low pressure conditions, annulus pressure exceeding MAASP, sustained casing pressure, unsuccessful bleed-down behaviour, rapid pressure change, and abnormal post-bleed-down pressure build-up. The system also considers the operational state of the well. Changes caused by deliberate operator actions, such as choke adjustments, are identified so that expected pressure responses can be distinguished from unexpected abnormalities. This helps reduce unnecessary alarms and improves the usefulness of the monitoring system. The dashboard provides colour-coded well tiles for rapid identification of well status and allows users to drill down into tubing, casing, and annulus pressure trends. An integrity watch-list highlights wells requiring attention, while production-engineering panels provide information related to pressure decline, drawdown, blockage, sand-up, and possible hydrate-related production problems. Alarm escalation is separated according to severity. Warning-level conditions are directed towards field operators, while critical integrity conditions are escalated to the well-integrity engineer and HSE personnel. The overall system provides a structured method for real-time well surveillance, pressure monitoring, alarm management, and integrity-focused decision support. The project demonstrates how industrial sensor data, rule-based engineering logic, well-specific configuration, real-time visualization, and edge-based monitoring can be integrated into a unified application. The system provides a practical foundation for improving well surveillance, reducing hazardous manual gauge-reading activities, supporting early detection of integrity risks, and improving operational response.

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1. ORGANIZATION AND INDUSTRY EXPOSURE

1.1 Organization Profile

The internship provided practical exposure to industrial software development, edge computing, real-time monitoring, and oil and gas operational technology. The project was developed in the context of industrial monitoring, where continuous acquisition and analysis of field sensor data are important for maintaining safe and reliable operations. The project domain combines Industrial IoT, Edge Computing, Real-Time Monitoring, Oil and Gas Well Surveillance, Pressure Monitoring, Alarm Management, and Well Integrity.

The Wellhead Pressure Monitoring Dashboard focuses on converting continuous pressure measurements from wellhead instrumentation into meaningful operational and integrity information. The system is intended to support field operators, production engineers, well-integrity engineers, and HSE personnel by presenting important well conditions through a centralized monitoring interface.

1.2 Internship Role and Responsibilities

During the internship, I worked on the development of the **Wellhead Pressure Monitoring Dashboard**. My responsibilities involved understanding the well-monitoring requirements, analysing wellhead pressure parameters, designing the monitoring workflow, implementing rule-based pressure checks, configuring well-specific limits, developing dashboard requirements, testing abnormal conditions, and supporting deployment and verification.

Responsibilities

- Studied the fundamentals of wellhead pressure monitoring and well integrity.
- Analysed tubing, casing, annulus, flowline, and choke pressure parameters.
- Understood the role of RTUs in acquiring wellhead pressure measurements.
- Designed a normalized and well-tagged pressure-data workflow.
- Implemented sensor-data validation for range, calibration drift, and stuck readings.
- Separated instrument faults from process and integrity alarms.
- Developed a per-well YAML configuration structure.
- Configured MAASP values for accessible casing annuli.
- Configured tubing MAOP and normal operating pressure bands.
- Configured expected shut-in pressure for individual wells.
- Implemented tubing HI/HIHI and LO/LOLO pressure rules.
- Implemented annulus pressure-above-MAASP monitoring.
- Implemented sustained casing pressure detection.
- Implemented pressure bleed-down monitoring.
- Implemented pressure build-up rate analysis after bleed-down.
- Used dP/dt analysis to identify rapid pressure changes.

- Considered well-state changes and operator actions during alarm evaluation.
- Designed colour-coded well status indicators.
- Developed requirements for tubing, casing, and annulus trend visualization.
- Designed an integrity watch-list for critical wells.
- Included production-engineering indicators such as drawdown and pressure decline.
- Considered production-loss conditions such as blockage, sand-up, and hydrate formation.
- Designed alarm escalation according to severity.
- Performed functional and integration testing of pressure-monitoring conditions.
- Verified dashboard behaviour and alarm presentation.
- Supported application deployment and monitoring in the available edge environment.

1.3 Faculty Mentor Interaction

Regular interaction was maintained with the faculty mentor throughout the internship. Project progress, implementation activities, technical challenges, testing results, and deployment-related activities were discussed periodically. The mentor feedback helped in improving the technical approach, organizing the project into manageable modules, and maintaining alignment between the project objectives and the implemented monitoring functionality.

1.4 Industry Practices and Processes Observed

The following industrial software and engineering practices were considered during the project:

- **Requirement Analysis :**The well-monitoring requirements were analysed and divided into data acquisition, validation, pressure-rule evaluation, alarm management, visualization, and escalation activities.
- **Sensor Data Validation** : Incoming pressure measurements require validation before they are used for alarm generation. Range validation, calibration-drift checks, and stuck-sensor detection are used to distinguish reliable measurements from instrument problems.
- **Configuration-Driven Monitoring :** Well-specific parameters are maintained separately through YAML configuration. This allows different wells to have different MAASP, MAOP, operating bands, annulus availability, and expected shut-in pressure values.
- **Rule-Based Engineering :** Pressure conditions are evaluated using predefined engineering rules. This provides transparent and explainable alarm conditions that can be understood by operations and integrity personnel.
- **Alarm Management** : Alarm severity is separated according to operational importance. Warning conditions are directed to field operators, while critical integrity conditions are escalated to well-integrity and HSE personnel.

- **Real-Time Monitoring** : Pressure measurements are continuously processed so that abnormal conditions can be identified without requiring personnel to manually visit individual wells.
- **Testing** : The system requires functional testing of individual rules, integration testing of the complete pressure-monitoring workflow, and dashboard testing to verify that alarms and well states are displayed correctly.
- **Documentation** : Well configurations, monitoring rules, system architecture, test cases, results, and deployment activities are documented to support maintainability and future improvements.

1.5 Tools, Technologies and Platforms

The following technologies/components are relevant to the Wellhead Pressure Monitoring Dashboard based on the defined project architecture:

Category	**Tool / Technology**	**Purpose**
Data Acquisition	Wellhead RTU	Acquisition of field pressure measurements
Instrumentation	Pressure Transmitters	Measurement of tubing, casing, annulus, flowline, and choke pressures
Configuration	YAML	Well-specific configuration and engineering limits
Monitoring	Rule Engine	Evaluation of pressure and integrity conditions
Data Processing	Pressure Time-Series Processing	Processing normalized, well-tagged measurements
Analysis	dP/dt Calculation	Detection of rapid pressure changes
Visualization	Web Dashboard	Live well monitoring and trend visualization
Alarm Management	Severity-Based Alarm Logic	Warning and critical alarm escalation
Containerization	Docker	Packaging the monitoring application
Edge Computing	EVE-OS	Edge execution environment
Edge Management	ZEDEDA	Edge-node and application management
Virtualization	Oracle VirtualBox	Virtual edge-node environment

Development	VS Code	Application development and debugging
Version Control	Git	Source-code version management

CO Alignment: CO1 – Industry Practices, Processes, Techniques and Technology.

The internship provided practical understanding of industrial monitoring requirements, sensor-data handling, rule-based engineering logic, dashboard development, alarm management, edge deployment, testing, and technical documentation.

2. INTERNSHIP OBJECTIVES AND LEARNING TARGETS

The main objectives of the internship were focused on understanding industrial well monitoring, developing a real-time pressure-monitoring solution, implementing integrity-focused alarm rules, and gaining practical experience in industrial software deployment.

O1 – Understand Wellhead Pressure Monitoring
Understand the purpose of tubing, casing, annulus, flowline, and choke pressure monitoring and their importance in well operations and integrity management.

O2 – Analyse Well Pressure Requirements
Analyse well-specific operating limits such as MAASP, tubing MAOP, normal operating bands, and expected shut-in pressures.

O3 – Develop a Rule-Based Monitoring System
Develop pressure-monitoring rules capable of detecting abnormal pressure conditions, sustained casing pressure, bleed-down failure, rapid pressure changes, and abnormal pressure build-up.

O4 – Develop a Centralized Dashboard
Provide a single monitoring interface containing well status, pressure trends, integrity alerts, and production-related information.

O5 – Implement Alarm Escalation
Separate operational warnings from critical integrity alarms and support appropriate escalation to field operators, well-integrity engineers, and HSE.

O6 – Test and Deploy the Application
Perform functional, integration, dashboard, alarm, and deployment testing and verify the monitoring workflow in the available edge environment.

3. PROBLEM / REQUIREMENT IDENTIFICATION AND ANALYSIS

3.1 Background and Context

Oil and gas wells contain several pressure points that need to be monitored continuously for safe and efficient operation. Tubing pressure, casing pressure, annulus pressure, flowline pressure, and choke pressure provide important information about the operating condition of a well. Traditional field surveillance may require personnel to travel from one well to another to manually read gauges. This approach can result in delayed detection of abnormal conditions and can expose personnel to unnecessary operational hazards.

Pressure abnormalities may also develop gradually and may not always be identified by a single fixed pressure threshold. For example, sustained casing pressure, abnormal annulus build-up, unsuccessful bleed-down, or an unusual post-bleed-down pressure recovery can provide important indicators of well-integrity concerns.Therefore, there is a need for a centralized monitoring solution that can continuously collect pressure information, apply well-specific engineering rules, identify abnormal conditions, and present the results through an easy-to-understand dashboard.

3.2 Problem Statement

The project addresses the problem of continuous and centralized monitoring of wellhead pressure conditions across multiple wells.

The proposed system should:

- Collect tubing-head pressure measurements.
- Collect accessible casing annulus A/B/C pressures.
- Collect flowline pressure.
- Collect choke pressure.
- Process measurements at approximately 1–10 second intervals.
- Associate every measurement with the correct well.
- Validate sensor measurements.
- Identify instrument faults separately from process alarms.
- Compare annulus pressure against MAASP.
- Compare tubing pressure against configured operating limits.
- Detect sustained casing pressure.
- Detect unsuccessful bleed-down behaviour.
- Calculate pressure build-up rates.
- Identify rapid pressure changes using dP/dt.
- Identify production-loss indicators such as blockage, sand-up, and hydrate formation.
- Suppress or explain alarms caused by deliberate operator actions.
- Provide a centralized live dashboard.
- Provide severity-based alarm escalation.

3.3 Existing Process / Existing System

In a conventional well-surveillance process, field personnel may depend on local gauges and periodic manual inspection to understand well pressure conditions. Such an approach can make continuous monitoring difficult when a large number of wells are distributed across a field. Manual monitoring can also delay the identification of pressure build-up or abnormal changes because the measurement is only available when the well is physically inspected.

Another limitation is that individual pressure readings may not provide sufficient context. A pressure value needs to be interpreted according to the well's completion configuration, MAASP, MAOP, normal operating range, expected shut-in condition, and current operating state.These limitations create a need for a centralized, automated, rule-based monitoring system.

3.4 Requirements and Constraints

Type	**Requirement / Constraint**	**Priority**
Functional	Acquire tubing, casing, annulus, flowline, and choke pressures	High
Functional	Process normalized, well-tagged readings	High
Functional	Validate sensor measurements	High
Functional	Detect tubing HI/HIHI and LO/LOLO conditions	High
Functional	Detect annulus pressure above MAASP	Critical
Functional	Detect sustained casing pressure	High
Functional	Detect unsuccessful pressure bleed-down	High
Functional	Calculate pressure build-up rate	High
Functional	Detect rapid dP/dt conditions	High
Functional	Display pressure trends	High
Functional	Provide integrity watch-list	High
Functional	Provide production monitoring indicators	Medium
Non-functional	Provide timely alarm generation	High
Non-functional	Reduce unnecessary alarms	High
Non-functional	Provide clear and understandable visualization	High

Technical	Support well-specific YAML configuration	High
Technical	Support continuous time-series pressure data	High
Technical	Support configurable engineering limits	High
Operational	Support monitoring without manual well-to-well visits	High
Operational	Provide severity-based escalation	High

3.5 Problem Analysis

The primary challenge is to distinguish normal pressure behaviour from abnormal conditions across multiple wells. Different wells may have different completion configurations, available annuli, MAASP values, tubing MAOP values, normal operating pressure bands, and expected shut-in pressures. Therefore, a single fixed threshold cannot be applied to every well.

The monitoring system must also distinguish between:

1. Normal operating changes
2. Deliberate operator actions
3. Sensor/instrument faults
4. Production-related abnormalities
5. Well-integrity abnormalities

A further challenge is that some integrity conditions are better identified through pressure behaviour over time rather than a single pressure value. Sustained casing pressure, bleed-down failure, and post-bleed-down pressure build-up are examples where the trend and rate of change are important.

CO Alignment: CO2 – Analyze and design solutions to complex business problems.

4. SOLUTION DESIGN AND METHODOLOGY

4.1 Proposed Solution

The proposed **Wellhead Pressure Monitoring Dashboard** is a centralized real-time monitoring system designed to continuously observe pressure conditions across the well stock. Pressure transmitters located at the tubing head, accessible casing annuli, flowline, and choke provide measurements to wellhead RTUs. The measurements are acquired at short intervals and published as normalized, well-tagged readings. The incoming data passes through validation checks to identify invalid values, calibration drift, and stuck sensors. Instrument faults are classified separately so that an unreliable measurement is not incorrectly interpreted as an actual well-integrity event. A well-specific YAML configuration register provides the engineering

information required by the monitoring rules. The configuration contains the completion type, available annuli, MAASP values, tubing MAOP, normal operating bands, and expected shut-in pressure. The rule engine continuously compares incoming pressure data with these configured parameters. It identifies high and low tubing pressure conditions, annulus pressure above MAASP, sustained casing pressure, unsuccessful bleed-down behaviour, rapid pressure changes, and abnormal pressure build-up.

The system also considers the current well state and deliberate operator actions. For example, a planned choke adjustment may naturally produce a pressure change. Such expected behaviour should be recognized so that unnecessary alarms are suppressed or appropriately explained. The processed information is displayed through a dashboard containing colour-coded well tiles, pressure trends, an integrity watch-list, and production-engineering panels. Overall, the proposed system combines real-time sensor acquisition, validation, configuration-driven rule processing, pressure trend analysis, alarm management, and dashboard visualization.

4.2 Major Components of System Architecture

1. Pressure Sensor Layer

Pressure transmitters provide measurements from:

- Tubing head
- Casing annulus A
- Casing annulus B
- Casing annulus C
- Flowline
- Choke

2. Wellhead RTU Layer

The wellhead RTU collects pressure measurements from the field instrumentation at intervals of approximately 1–10 seconds.

3. Data Normalization Layer

Incoming readings are normalized into a consistent format and associated with the corresponding well identifier.

4. Sensor Validation Layer

The validation component checks:

- Measurement range
- Calibration drift
- Stuck sensor behaviour
- Missing or invalid measurements
- Instrument faults

5. Well Configuration Layer

A YAML register stores well-specific information such as:

- Completion type
- Available annuli
- MAASP per annulus
- Tubing MAOP
- Normal operating band
- Expected shut-in pressure

6. Rule Engine

The rule engine evaluates pressure readings against configured limits and operating conditions.

7. Pressure Trend Analysis

Historical and current pressure values are analysed to identify:

- Pressure increase
- Pressure decline
- dP/dt
- Drawdown
- Pressure build-up
- Post-bleed-down recovery

8. Well-State Detection

The system identifies changes in well operating state and considers deliberate operator actions during alarm evaluation.

9. Alarm Management

Detected conditions are classified according to severity and alarm type.

10. Dashboard

The dashboard provides:

- Colour-coded well tiles
- Live pressure status
- Tubing trends
- Casing trends
- Annulus trends
- Integrity watch-list
- Drawdown information
- Pressure decline information
- Production-loss indicators

11. Escalation Layer
Warnings are communicated to field operators, while critical integrity conditions are escalated to well-integrity engineers and HSE personnel.

4.3 Module / Component Design

Module	Purpose	Inputs	Processing	Outputs
Module 1 – Pressure Data Acquisition	Collect wellhead pressure data	RTU sensor readings	Receive and tag measurements	Well-tagged pressure data
Module 2 – Data Validation	Ensure data quality	Pressure measurements	Range, drift, and stuck-sensor checks	Validated data / instrument alarm
Module 3 – Well Configuration	Provide well-specific limits	YAML register	Load MAASP, MAOP, bands, well state parameters	Configuration values
Module 4 – Pressure Rule Engine	Identify abnormal pressure	Validated pressure + configuration	Apply HI/HIHI, LO/LOLO and MAASP rules	Pressure alarms
Module 5 – Integrity Monitoring	Identify integrity risks	Annulus/casing trends	Evaluate sustained pressure, bleed-down and build-up	Integrity alerts
Module 6 – Rate Analysis	Detect rapid pressure changes	Pressure time series	Calculate dP/dt and pressure trends	Rapid-change indicators

Module 7 – Well-State Analysis	Identify expected behaviour	Pressure + operator action	Classify operating state	Alarm suppression/explanation
Module 8 – Production Monitoring	Identify production issues	Tubing/flowline/choke pressure	Analyse drawdown and pressure decline	Production indicators
Module 9 – Dashboard	Visualize well condition	Processed results	Display status, trends and alerts	Monitoring interface
Module 10 – Alarm Escalation	Direct alerts to appropriate users	Alarm severity	Apply escalation rules	Operator / integrity / HSE alerts

4.4 Design Decisions and Justification

Well-Specific Configuration

A YAML configuration register was selected because every well may have different pressure limits and completion characteristics. Configuration-driven monitoring makes the system easier to maintain and extend.

Rule-Based Monitoring

Rule-based logic was selected because pressure-integrity conditions need to be transparent and explainable. Engineers can understand why a particular alarm was generated.

MAASP-Based Monitoring

MAASP is used as an important reference for annulus pressure monitoring. Pressure exceeding the configured MAASP is treated as an integrity-focused condition.

Trend-Based Analysis

Some abnormal conditions cannot be reliably identified using a single measurement. Therefore, pressure trends, pressure build-up, bleed-down behaviour, and dP/dt are considered.

Instrument Fault Separation

Sensor faults are maintained as a separate alarm class. This prevents a faulty instrument from being incorrectly interpreted as a physical well-integrity problem.

Operator-Action Awareness
Planned operational changes such as choke adjustments can naturally change well pressure. Considering operator actions helps reduce unnecessary alarms.

Centralized Dashboard
A centralized dashboard eliminates the need to manually inspect every well and provides a common operating picture for multiple users.

Severity-Based Escalation
Different users require different levels of information. Operational warnings can be handled by field operators, while critical integrity conditions require escalation to well-integrity and HSE personnel.

CO Alignment: CO2 – Solution analysis, design and technical justification.

5. IMPLEMENTATION AND TECHNICAL EXECUTION

5.1 Development Environment

The system was designed as a real-time industrial monitoring application. The development workflow consists of pressure-data acquisition, data normalization, sensor validation, configuration loading, rule evaluation, pressure trend analysis, dashboard visualization, and alarm management. The well-specific YAML register acts as the configuration source for engineering limits. The monitoring logic uses these parameters to evaluate pressure conditions for each well.

The application can be containerized using Docker and deployed in an edge environment using EVE-OS and ZEDEDA, supporting localized processing and centralized edge management. Development, debugging, testing, and configuration activities are performed using the available software development environment.

5.2 Implementation Details

The implementation begins with pressure-data acquisition from the wellhead RTUs. Measurements from the tubing head, casing annuli, flowline, and choke are collected at short intervals. Each measurement is associated with the corresponding well identifier and normalized into a consistent format. This enables the monitoring system to process data from multiple wells using a common workflow. The validation module checks whether measurements fall within valid ranges and identifies abnormal instrument behaviour such as calibration drift and stuck sensors. Instrument faults are reported separately from process alarms. The YAML configuration module provides well-specific engineering parameters. The rule engine uses these values to determine whether a pressure condition is normal, warning-level, or critical.

Tubing pressure is evaluated against HI, HIHI, LO, and LOLO limits. Annulus pressure is compared against the configured MAASP. Sustained casing pressure and pressure build-up behaviour are analysed over time. For bleed-down monitoring, the system observes the pressure response after the bleed-down operation. If the pressure does not decrease as expected or subsequently builds up beyond the acceptable envelope, an integrity condition can be generated. The pressure-rate module evaluates dP/dt to identify rapid pressure changes. This can provide an additional indication of abnormal events such as leakage or slug-related behaviour.

The system also monitors production-related pressure behaviour. Drawdown and pressure decline can be presented to production engineers, while abnormal pressure behaviour can provide supporting information for possible blockage, sand-up, or hydrate-related production problems. The processed results are presented through the dashboard. Each well is represented using a colour-coded status tile. Users can select a well and view detailed tubing, casing, and annulus pressure trends. An integrity watch-list highlights wells requiring attention, while alarm severity determines the appropriate escalation path.

5.3 Individual Contribution

Task / Module	**My Contribution**	**Tools / Technologies**	**Output**
Pressure Data Acquisition	Worked on the pressure-data acquisition and well-tagging workflow	RTU, pressure transmitters	Normalized well pressure data
Sensor Validation	Designed validation checks for range, calibration drift, and stuck sensors	Rule-based validation	Validated readings / instrument alarms
YAML Configuration	Defined well-specific configuration parameters	YAML	Well configuration register
Pressure Rules	Implemented tubing and annulus pressure rules	Rule engine	Pressure alarms
Integrity Monitoring	Developed logic for MAASP, sustained casing pressure and bleed-down conditions	Pressure trend analysis	Integrity alerts
dP/dt Analysis	Analysed pressure rate of change	Time-series pressure data	Rapid-change indicators

Well-State Analysis	Considered operator actions and well-state changes	State/rule logic	Reduced false alarms
Dashboard	Designed monitoring views and well status visualization	Web dashboard	Live monitoring interface
Production Monitoring	Included drawdown and pressure-decline indicators	Pressure analysis	Production indicators
Alarm Escalation	Defined severity-based alarm handling	Alarm logic	Operator/integrity/HSE escalation
Deployment	Supported application containerization and edge deployment	Docker, EVE-OS, ZEDEDA	Deployed application
Testing	Tested pressure rules, alarms and dashboard behaviour	Test cases	Test results

5.4 Integration / Collaboration

The different modules were integrated to create a complete wellhead pressure monitoring workflow.

The overall sequence is:

Pressure Transmitters → Wellhead RTU → Data Normalization → Sensor Validation → YAML Configuration → Rule Engine → Pressure/Integrity Analysis → Alarm Management → Dashboard → Escalation

The integration ensures that raw pressure measurements are converted into useful operational and integrity information. The dashboard acts as the central interface between the monitoring system and its users. It provides current well status, pressure trends, active alarms, integrity conditions, and production-related information. The system is designed to support multiple wells and different well configurations without requiring the monitoring logic to be rewritten for every well.

CO Alignment: CO3 – Build and deploy solutions for the target platform.

6. TESTING, RESULTS AND DEPLOYMENT

6.1 Testing Strategy

Testing was designed to verify the correctness of pressure acquisition, validation, rule evaluation, alarm generation, dashboard visualization, and deployment.

The testing strategy includes:

- Sensor-data validation testing
- Configuration testing
- Pressure-limit testing
- MAASP testing
- Sustained casing pressure testing
- Bleed-down testing
- dP/dt testing
- Well-state testing
- Alarm escalation testing
- Dashboard testing
- Integration testing
- Deployment testing

6.2 Functional and Integration Test Cases

Test Case	Input / Scenario	Expected Result	Status
TC01	Valid tubing pressure	Reading accepted and displayed	Pass
TC02	Tubing pressure above HI limit	Warning generated	Pass
TC03	Tubing pressure above HIHI limit	Critical/high-severity alarm generated	Pass
TC04	Tubing pressure below LO limit	Low-pressure warning generated	Pass
TC05	Tubing pressure below LOLO limit	Critical low-pressure alarm generated	Pass
TC06	Annulus pressure exceeds MAASP	Integrity alarm generated	Pass
TC07	Pressure transmitter becomes stuck	Instrument-fault alarm generated	Pass
TC08	Calibration drift condition	Instrument condition identified	Pass
TC09	Sustained casing pressure	Integrity watch condition generated	Pass
TC10	Successful bleed-down	No integrity escalation	Pass
TC11	Failed bleed-down	Integrity alert generated	Pass

TC12	Rapid pressure change	dP/dt condition detected	Pass
TC13	Abnormal post-bleed-down build-up	Integrity condition detected	Pass
TC14	Planned choke change	Expected pressure response recognized	Pass
TC15	Normal well operation	No unnecessary alarm generated	Pass
TC16	Dashboard well selection	Detailed pressure trends displayed	Pass
TC17	Critical integrity condition	Escalation directed to appropriate users	Pass
TC18	Multiple wells operating simultaneously	Individual well states displayed correctly	Pass

6.3 Results and Performance

The developed system provides a centralized approach to monitoring wellhead pressure conditions. The dashboard allows users to observe multiple wells from a single interface rather than depending entirely on manual gauge readings.

The implementation provides:

- Continuous pressure monitoring
- Well-specific pressure limits
- Sensor-fault identification
- MAASP-based integrity monitoring
- Tubing pressure alarms
- Sustained casing pressure detection
- Bleed-down monitoring
- Pressure build-up analysis
- dP/dt monitoring
- Well-state awareness
- Pressure trend visualization
- Integrity watch-list
- Production monitoring indicators
- Severity-based alarm escalation

The system improves the organization of well-pressure information and provides an operationally useful method for identifying wells requiring attention. The use of well-specific configuration also

allows the same monitoring framework to support different wells with different completion types and pressure limits.

7. INDUSTRY / PROFESSIONAL LEARNING

7.1 Technical Learning

The internship provided practical understanding of:

- Wellhead pressure monitoring
- Oil and gas well integrity
- Tubing and casing pressure behaviour
- Annulus pressure monitoring
- MAASP and MAOP concepts
- RTU-based industrial data acquisition
- Real-time monitoring
- Time-series pressure analysis
- Rule-based alarm generation
- Sensor validation
- Dashboard development
- Edge computing
- Containerized deployment
- Industrial alarm management

7.2 Problem-Solving Skills

The project required analysing pressure behaviour rather than depending only on individual threshold values. This improved understanding of how engineering conditions can be converted into software rules. The project also demonstrated the importance of separating sensor faults from actual process abnormalities and considering operator actions before generating critical alarms.

7.3 Professional Skills

The internship strengthened skills in:

- Requirement analysis
- System design
- Technical documentation
- Testing
- Debugging
- Configuration management
- Dashboard design
- Industrial problem analysis
- Technical communication
- Project presentation

7.4 Industry Insights

The project provided practical understanding that industrial monitoring systems require more than simply displaying sensor values. A useful monitoring system must provide context, validation, alarm prioritization, historical trends, and actionable information. The project also demonstrated the importance of reducing unnecessary alarms. Excessive alarms can make it difficult for operators to identify the most important events. Therefore, operator-action awareness and alarm severity classification are important parts of the system design.

8. INTERNSHIP ACTIVITY LOG

Week	Activity / Task	Technical Learning	Challenge	Outcome
Week 1	Studied wellhead pressure monitoring and project requirements	Well integrity, pressure parameters, RTUs	Understanding different well pressure points	Initial architecture prepared
Week 2	Designed pressure-data acquisition and validation	Sensor validation and well-tagging	Handling abnormal sensor readings	Validation workflow developed
Week 3	Developed YAML well configuration	MAASP, MAOP, operating bands	Supporting different well configurations	Configuration register prepared
Week 4	Implemented pressure and integrity rules	HI/HIHI, LO/LOLO, MAASP, bleed-down	Avoiding false alarms	Rule engine developed
Week 5	Implemented pressure trends and well-state logic	dP/dt, sustained pressure, operator actions	Distinguishing expected and unexpected changes	Trend and state monitoring implemented
Week 6	Developed dashboard, testing and deployment	Visualization, alarm escalation, deployment	Integration and testing issues	Monitoring dashboard and test results completed

9. EVIDENCE OF INTERNSHIP WORK

The major evidence of the project work can include the following figures and screenshots:

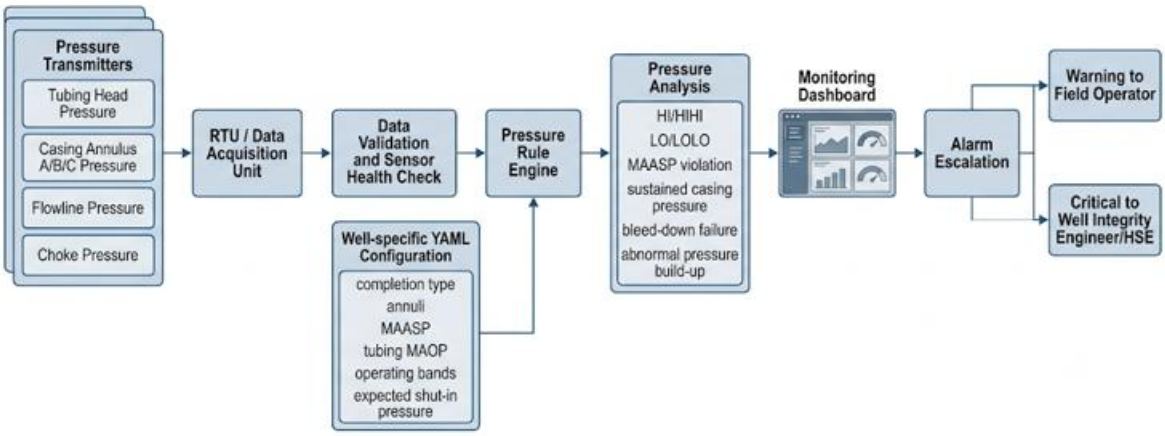


Fig. 9.1 – Overall System Architecture

Wellhead Pressure Monitoring Dashboard architecture showing pressure transmitters, RTU data acquisition, data validation, YAML configuration, rule engine, pressure analysis, dashboard, and alarm escalation.

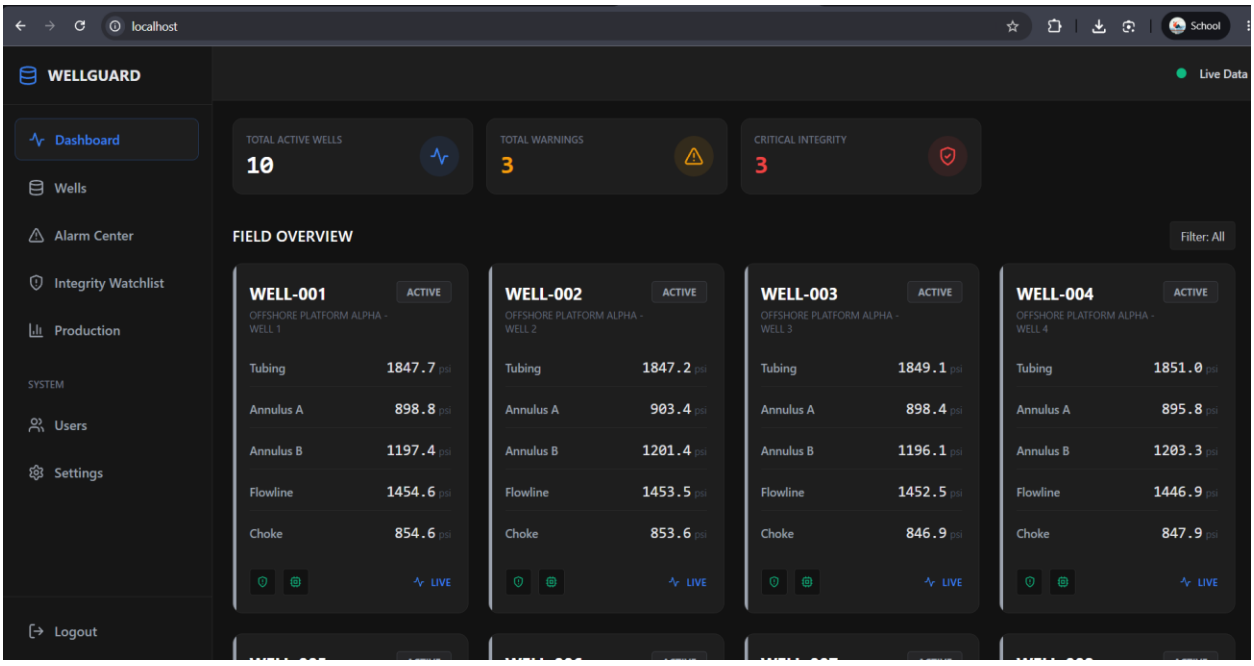


Fig. 9.2 – Well Monitoring Dashboard Dashboard showing colour-coded well tiles, current pressure conditions, active alarms, and overall well status.

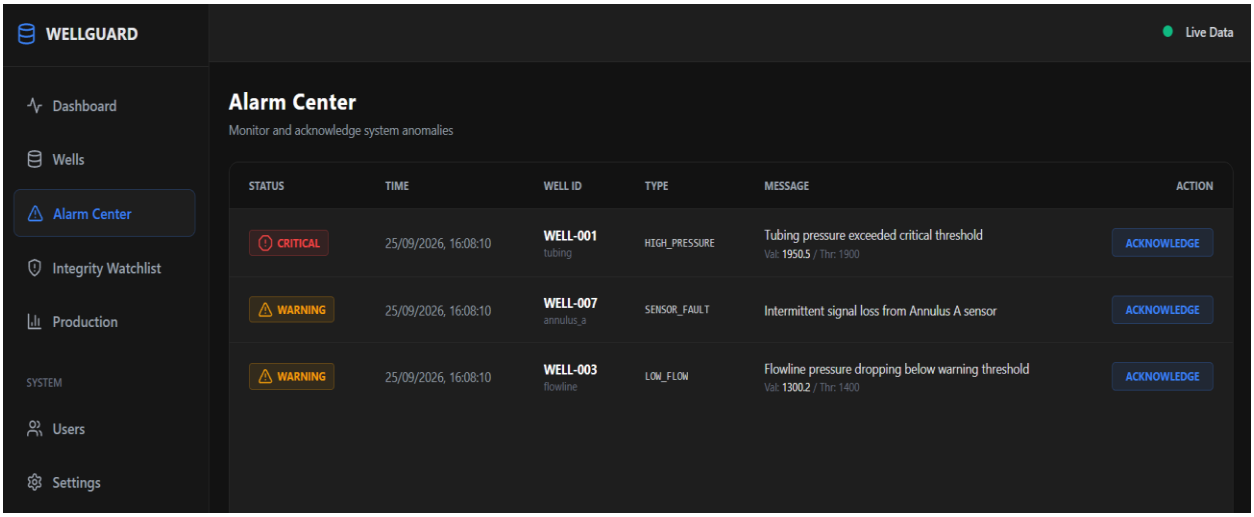


Fig. 9.3 – Alarm Center Tubing, casing, and annulus pressure trends showing normal and abnormal pressure behaviour over time.

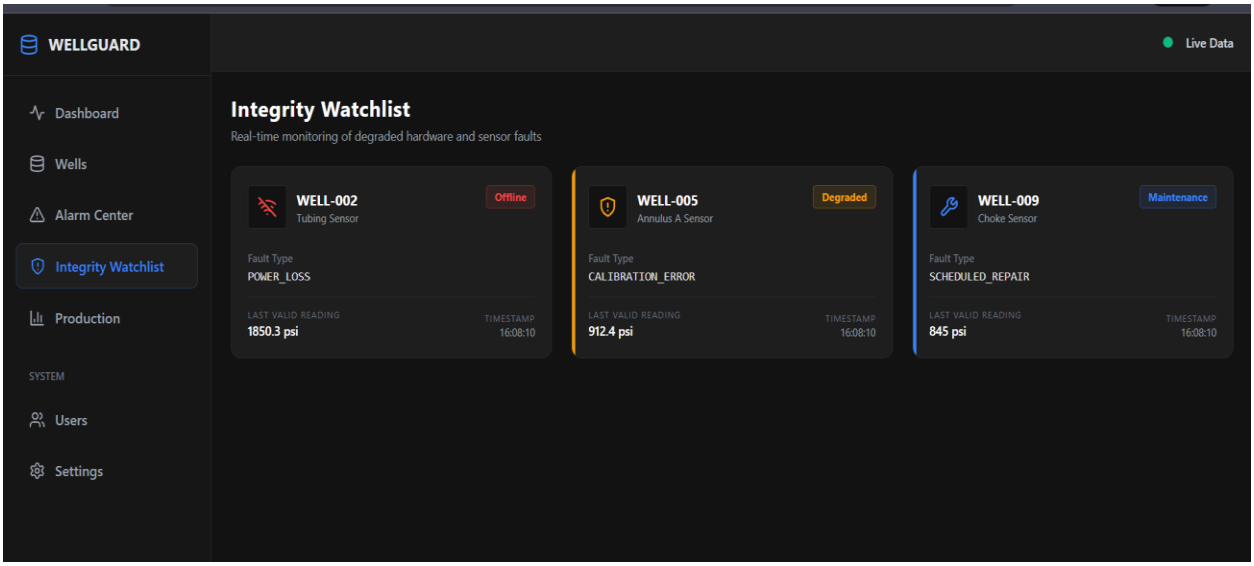


Fig. 9.4 – Integrity Watch-List Integrity watch-list showing wells with sustained casing pressure, MAASP violations, bleed-down failure, or abnormal pressure build-up.

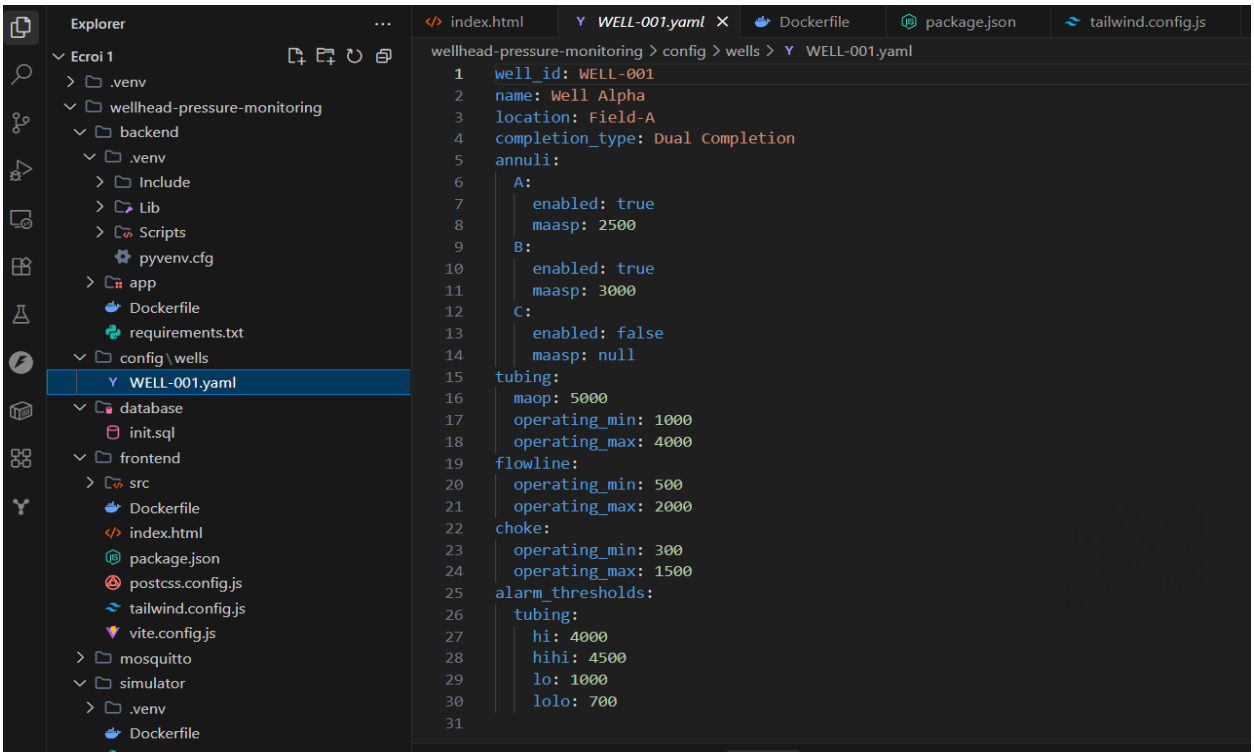


Fig. 9.5 – YAML Configuration Well-specific YAML configuration showing completion type, annuli, MAASP, tubing MAOP, operating bands, and expected shut-in pressure.

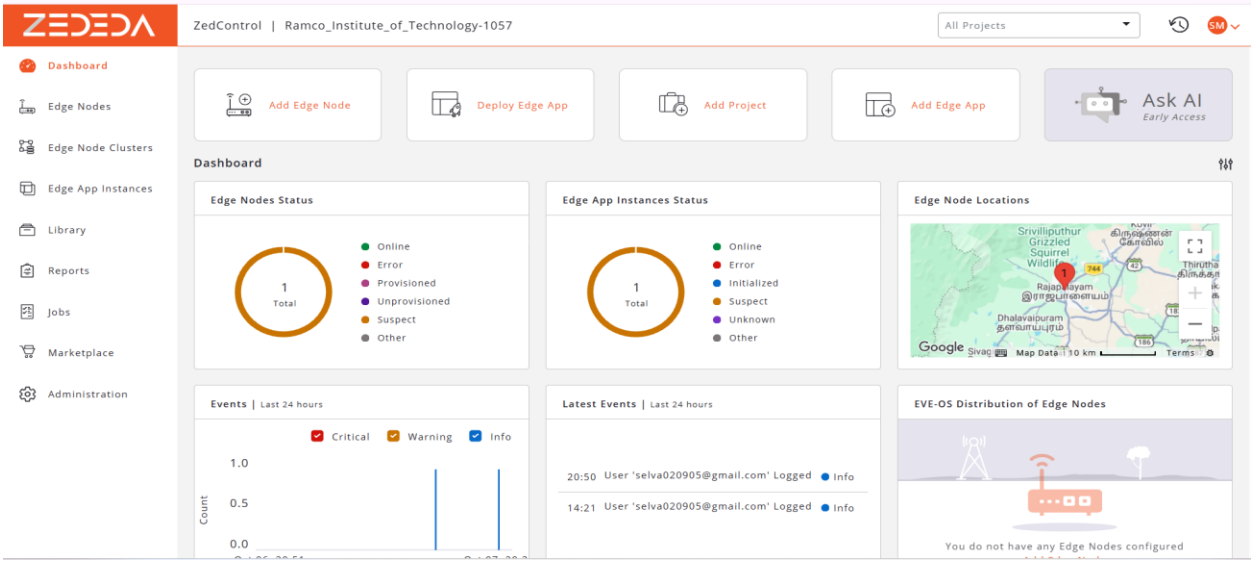


Fig. 9.6 – ZEDEDA edge deployment showing the EVE-OS edge node, containerized anomaly-detection application, application status, and deployment verification.

10. COURSE OUTCOME ATTAINMENT EVIDENCE

CO	Course Outcome	Evidence / Activities	Supporting Artifact
CO1	Industry Practices, Processes, Techniques and Technology	Requirement analysis, RTU monitoring, rule-based engineering, dashboard development, testing and deployment	Sections 1, 5, 6
CO2	Analyze and design solutions to complex business problems	Analysed manual well surveillance limitations and designed centralized pressure monitoring	Sections 3 and 4
CO3	Build and deploy solutions for the target platform	Implemented pressure monitoring, alarm rules, dashboard, containerization and edge deployment	Sections 5 and 6
CO4	Preparation of Technical Reports and Presentations	Prepared system documentation, test cases, technical analysis, presentation and final report	Sections 7–11

The project demonstrates the application of software engineering concepts to an industrial oil and gas monitoring problem. It connects requirement analysis, system design, implementation, testing, deployment, and documentation into a complete internship activity.

11. CONCLUSION

The internship project on **Wellhead Pressure Monitoring Dashboard** provided practical exposure to industrial monitoring, well integrity, real-time sensor-data processing, rule-based alarm management, dashboard development, and edge computing. The developed concept provides a centralized live view of tubing, casing, annulus, flowline, and choke pressures across multiple wells. By combining normalized well-tagged pressure measurements with well-specific configuration parameters, the system can evaluate pressure conditions according to the operating characteristics of each well.

The monitoring rules support detection of tubing HI/HIHI and LO/LOLO conditions, annulus pressure above MAASP, sustained casing pressure, unsuccessful bleed-down, abnormal pressure build-up, and rapid pressure changes. The system also considers deliberate operator actions to reduce unnecessary alarms and separates instrument faults from actual well-integrity conditions. The dashboard provides colour-coded well status, pressure trends, an integrity watch-list, and production-related indicators such as drawdown and pressure decline. Severity-based escalation ensures that operational warnings can be directed to field operators while critical integrity conditions can be escalated to well-integrity engineers and HSE personnel.

Overall, the project demonstrates how real-time industrial data and engineering rules can be integrated into a centralized monitoring platform. The proposed system can reduce dependence on manual well-to-well gauge inspection, support earlier identification of abnormal well conditions, improve situational awareness, and provide a structured foundation for safer and more efficient well surveillance. The internship strengthened practical skills in requirement analysis, system architecture, sensor-data validation, rule-based monitoring, dashboard design, testing, deployment, technical documentation, and industrial problem-solving.

12. REFERENCES / RESOURCES

[1] R. Rocha-Valadez, A. K. Wojtanowicz, and R. A. A. Al-Majed, **"Inherently safer sustained casing pressure testing for well integrity evaluation,"** *Journal of Loss Prevention in the Process Industries*, vol. 29, pp. 209–215, 2014. https://doi.org/10.1016/j.jlp.2014.02.012

[2] R. Xu and A. K. Wojtanowicz, **"Pressure buildup test analysis in wells with sustained casing pressure,"** *Journal of Natural Gas Science and Engineering*, vol. 38, pp. 608–620, 2017. https://doi.org/10.1016/j.jngse.2016.12.033

[3] S. Wu, L. Zhang, J. Fan, X. Zhang, D. Liu, and D. Wang, **"A leakage diagnosis testing model for gas wells with sustained casing pressure from offshore platform,"** *Journal of Natural Gas Science and Engineering*, vol. 55, pp. 276–287, 2018. https://doi.org/10.1016/j.jngse.2018.05.006

[4] Y. Wang, J. Zeng, and D. Gao, **"Assessment on the sustained casing pressure (SCP) with consideration of the infiltration of formation fluid into the annulus of deepwater wells,"** *Journal of Petroleum Science and Engineering*, vol. 189, 2020, Art. no. 106999. https://doi.org/10.1016/j.petrol.2020.106999

APPENDIX A – INTERNSHIP CERTIFICATE

[Attach internship completion certificate.]

APPENDIX B – OFFER / JOINING LETTER

[Attach offer/joining letter if applicable.]

APPENDIX C – INTERNSHIP PRESENTATION EVALUATION SCORE SHEET

S.No.	Evaluation Criterion	CO	Max. Marks	Marks Awarded
1	Industry Exposure & Domain Understanding	CO1	20	
2	Problem Identification & Analysis	CO2	15	
3	Solution Design & Approach	CO2	15	
4	Implementation & Technical Skills	CO3	15	
5	Testing, Results & Deployment	CO3	10	
6	Technical Report & Documentation	CO4	10	
7	Presentation & Communication Skills	CO4	10	
8	Viva / Response to Questions	CO1–CO4	5	
	TOTAL		100	

Overall Marks: _____ / 100

Evaluator Name & Signature: _____

Date: _____